

PRM393 – Final Assignment

Flutter Mobile Application Development

1. Objective

Develop a cross-platform mobile application using **Flutter** that demonstrates software engineering principles, mobile UI/UX design, cloud integration, and RESTful API consumption.

Students are free to choose any application domain (e.g., education, healthcare, e-commerce, social networking, productivity, tourism, entertainment, IoT, AI, etc.).

2. Technical Requirements

The project **must** satisfy the following requirements:

Mandatory Technologies

- Flutter (latest stable version)
- Dart
- Git & GitHub

Firebase Integration

Use **at least two (2)** Firebase services:

- Firebase Authentication
- Cloud Firestore
- Realtime Database
- Firebase Storage
- Firebase Cloud Messaging (FCM)
- Firebase Analytics
- Firebase Crashlytics
- Firebase Remote Config

REST API

Integrate at least **one (1)** RESTful API:

- Self-developed backend API, or
 - Public REST API
-

3. Minimum Functional Requirements

The application must include:

- User authentication
- User profile management
- CRUD operations
- REST API integration
- Firebase integration
- Search or filtering feature
- Input validation and exception handling
- Responsive and user-friendly interface

Students are encouraged to implement additional features beyond the minimum requirements.

4. Submission

Each team must submit:

- GitHub repository
 - APK file (Android)
 - Project report (PDF)
 - Presentation slides
-

5. Assessment Policy

Before the final evaluation, every project must satisfy the following conditions:

- The project must compile successfully.
- The source code will be reviewed before feature evaluation.
- The application **must be demonstrated on a physical Android or iOS device.**
- **Emulators or simulators are not allowed** during the final demonstration.
- Students must be able to explain their implementation and answer technical questions.

Projects that cannot be built or demonstrated on a physical device may receive significant score deductions.

6. Assessment Rubric (100 Points)

Category	Description	Points
Flutter Application Development	Project architecture, navigation, reusable widgets, responsive UI	20
Firebase Integration	Correct implementation of Firebase services	15

Category	Description	Points
REST API Integration	API consumption, asynchronous programming, error handling	15
Functional Completeness	Implementation of required features and overall functionality	20
UI/UX Design	User interface quality, usability, responsiveness	10
Code Quality	Clean architecture, coding standards, modularity, project organization, Git commit history	10
Innovation	Creativity, advanced features, technical complexity	5
Presentation & Live Demonstration	Professional presentation, successful demonstration on a physical mobile device, Q&A	5
Total		100

7. Academic Integrity

- All submitted work must be the students' own work.
- External libraries and public APIs may be used with proper attribution.
- AI-assisted development tools (e.g., ChatGPT, GitHub Copilot, Gemini, Cursor) are permitted. However, students are fully responsible for understanding and explaining every part of their implementation during the presentation and code review.
- Any plagiarism or inability to explain the submitted work may result in a failing grade.